Multi-View Feature Fusion and Visual Prompt for Remote Sensing Image Captioning

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***Abstract*— Remote sensing image (RSI) captioning is a vision-language multimodal task concentrating on both image comprehension and sentence generation. Several studies suggest that encoder–decoder-based methods have achieved success in RSI captioning. However, existing encoder–decoder-based methods may not fully explore image representations for RSI captioning and suffer from a lack of additional prompt information for sentence generation. In this article, a novel multi-view feature fusion and prompt (MVP)-based model is proposed to obtain better RSI representations and enhance language model performance in RSI captioning. Specifically, we design an attention-based feature fusion module to dynamically fuse multi-view visual features, which are extracted from the fine-tuned vision-language pretraining (VLP) model and the vision-task pretraining (VP) model. Then, a flexible visual prefix mapping module is proposed to transform images into visual prefixes, providing semantic information for the subsequent sentence generation. Finally, a BERT-based caption generator is applied to generate accurate descriptions based on the fused visual features and the visual prefixes, which are both outputs from our designed modules. Extensive experiments are conducted on three well-known benchmark datasets, demonstrating that our method achieves state-of-the-art (SOTA) performance. The relevant code is available at https://github.com/QiaoLing-Lin/MVP.**

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I. Introduction

REMOTE sensing image (RSI) captioning, a specified branch of image captioning, aims at automatically describing the contents of RSIs in natural language sentences. This technology can assist humans in conveniently comprehending the abundant geographic information contained in RSIs and offer practical applications across various domains, including urban planning, disaster monitoring, and military intelligence [1], [2].

In recent years, there has been a growing interest in RSI captioning methods based on encoder–decoder architectures [2], [3]. Compared with template-based methods [4] and retrieval-based methods [5], encoder–decoder-based methods perform better in generating flexible and diverse sentences, having become the premier methods in RSI captioning. Specifically, the encoder–decoder architecture is capable of translating RSIs directly into corresponding descriptions. It achieves this by utilizing image encoders to extract RSI features, followed by language decoders that process these features for sentence generation. To enhance RSI representations, RASG [6] adopts the pretrained VGG-16 model to extract visual features of RSI, while MLA [7] and Structure Attention [8] choose the pretrained ResNet model. For the choice of language decoders, variants of long short-term memory (LSTM) networks [6] and Transformer-based models [9] are welcomed to complete sentence generation. Despite their achievements, current methods have two limitations in terms of RSI representations and sentence generation.

Regarding RSI representations, current encoder–decoder-based methods may not fully explore image representations for RSI captioning. To learn RSI representations, current methods [6], [7], [8] mostly use the vision-task pretraining (VP) models, such as ResNet pretrained on the ImageNet dataset, to extract visual features. However, RSI captioning is a vision-language multimodal task. The visual features extracted from the VP model may achieve suboptimal performance for RSI captioning, encouraging us to turn to task-related models for guidance. Recently, there has been a growing emergence of large vision-language pretraining (VLP) models, catering to the demands of various multimodal tasks [10], [11], [12], [13]. Compared to VP models, it has been demonstrated that the visual features extracted from VLP models may contain more

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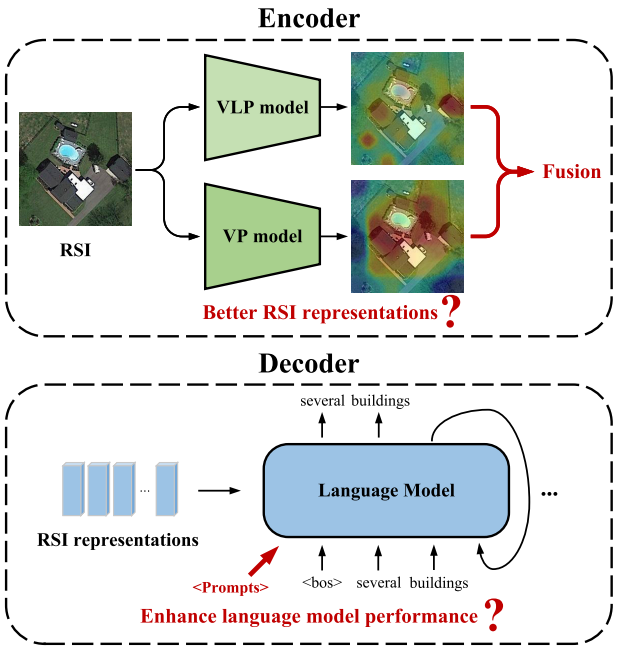


Fig. 1. Motivation of our research: to produce better RSI representations based on different visual features and enhance language model performance with prompt-based learning. In the encoder, it shows the visualization of the visual features extracted from the VLP model and the VP model. Obviously, the visual features extracted from VP models pay more attention to the entire RSI, while those from VLP models show interest in salient objects under the guidance of language. In the decoder, the language model usually starts sentence generation from a specified beginning token ⟨bos⟩. It is noticed that we denote the designed prompts as ⟨Prompts⟩ in the figure, which will be further described in the article.

aligned image-sentence knowledge, which is beneficial for downstream multimodal tasks [13]. Inspired by this, bringing VLP models into RSI captioning is worthy of exploration. Fig. 1 illustrates differences in the visual features extracted from the VLP model and the VP model, suggesting that these features contain rich information from different perspectives. Therefore, how to extract and fuse these multi-view visual features deserves further investigation.

When it comes to sentence generation, current encoder–decoder-based methods suffer from a lack of additional prompt information. Previous language decoders received the extracted RSI features before starting sentence generation from a specified beginning token. When encountering different RSIs, it is not enough for language decoders to generate sentences merely based on the identical beginning token. Furthermore, directly integrating language decoders in RSI captioning may be too coarse to learn a semantic mapping from the RSI domain to the language domain [6]. Therefore, the quest for additional information to guide RSI captioning language decoders becomes meaningful. A novel paradigm called prompt-based learning has recently gained attention in generative tasks [14], [15]. The core of this paradigm is to design appropriate prompts as additional information in order to enhance the performance of the pretrained model [16], [17], [18]. To the best of our knowledge, limited attention has been devoted to incorporating prompt-based learning into RSI captioning, offering a potential area for exploration. Fig. 1 provides an example of how prompt-based learning can be applied in RSI captioning.

In light of the mentioned limitations, we proposed a novel multi-view feature fusion and prompt (MVP)-based model, integrating a multi-view feature fusion (MVFF) module, a visual prefix mapping module, and a BERT-based caption generator for RSI captioning. Specifically, we designed an attention-based feature fusion module to fuse multi-view visual features as RSI representations. Then, a novel prompt-based framework was proposed to produce visual prefixes for the subsequent generation task. Besides, we employed a BERT-based model as the caption generator, which constitutes the entire RSI captioning architecture with the aforementioned modules.

Overall, the main contributions of our paper are summarized as follows.

1) To obtain better RSI representations, we designed an attention-based MVFF module to fuse two kinds of visual features. These visual features were extracted from the fine-tuned VLP model, contrastive language-image pretraining (CLIP), and the VP model, ResNet-101, respectively.
2) We proposed a flexible two-submodule framework, named visual prefix mapping, consisting of a visual feature extractor and a mapping network, to produce visual prefixes for sentence generation. Under the condition of visual prefixes, it may further explore the potential ability of the pretrained language model and enhance the entire RSI captioning model.
3) We proposed a novel prompt-based encoder–decoder architecture, which is composed of an MVFF module, a visual prefix mapping module, and a BERT-based caption generator for RSI captioning.
4) Extensive experiments have been conducted on three RSI captioning benchmark datasets. The results illustrate that our method can surpass several state-of-the-art (SOTA) methods, demonstrating the effectiveness and the superiority of the combinations of our designed modules.

The remainder of this article is organized as follows. Section II introduces some previous related studies on RSI captioning. Next, Section III describes the overall structure and module details of the proposed MVP model. Then, Section IV gives the experiments and analysis on three datasets. Finally, we conclude this study in Section V.

II. Related Work

Encoder–decoder-based methods have become the mainstream method in RSI captioning. Based on the encoder–decoder architecture, it can be improved from three perspectives: enhancing RSI representations, improving language decoders, and integrating approaches to both RSI representations and language decoders.

A. RSI Captioning Based on Enhanced Image Representations

Most methods apply attention mechanisms and feature fusion strategies to produce enhanced RSI representations,

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while grid features and local patch features give a new understanding of RSIs. To bridge the semantic gap between different-level features, Zhang et al. [19] introduced an attention mechanism based on the extracted attributes to further enhance the RSI representations. Taking into account the label information of RSI, Zhang et al. [20] proposed a label-attention-mechanism method for attending image features. These two attention mechanisms are both aimed at producing RSI representations containing rich information. The front one combines different-level features without multilabel training, while the latter utilizes label information from RSIs to calculate attention masks. Considering the noise of features from different layers, Huang et al. [21] proposed a denoising-based multiscale feature fusion mechanism to improve the ability of multiscale feature representations. Besides, grid-based visual features were introduced to improve RSI captioning performance by Zhuang et al. [22]. Visual Transformer [23] was brought into RSI captioning by Ren et al. [24], allowing local patch features to explore rich information of RSI.

B. RSI Captioning Based on Improved Language Decoders

Regarding language decoders, not only recurrent neural networks (RNNs) and LSTM are commonly used to achieve sentence generation, but Transformer-based decoders can produce high-quality sentences. To capture the long-term information, Wang et al. [25] introduced a novel retrieval topic recurrent memory network to utilize the topic words for sentence generation. In addition, Li et al. [7] proposed a multilevel attention model containing three attention structures in the decoder, which automatically focus on either the image or the sentence information when generating words. Considering knowledge in the vision domain, Zhao et al. [8] proposed a structured attention module to utilize the structured information of semantic contents in RSIs and facilitate interaction with the decoder during captioning. Besides, while Shen et al. [9] adopted modified Transformers as language decoders, Liu et al. [26] proposed multilayer aggregated Transformer, which regard LSTM as the connectivity to aggregate the features from Transformer encoders to generate sentences.

C. RSI Captioning Based on Integrated Approaches

Most encoder–decoder-based methods employ integrated approaches to improve both RSI representations and language decoders for better RSI captioning performance. Qu et al. [1] first proposed a multimodal neural network model with convolution neural networks (CNN) and RNN for high-resolution RSI captioning. Lu et al. [2] constructed a RSI captioning dataset (RSICD), and brought attention mechanisms into the CNN-RNN architecture. Moreover, Li et al. [6] introduced a recurrent attention mechanism to generate context-aware features for RSI representations, followed by the two-layer LSTM with the semantic gate to finally generate sentences. Considering multilabel semantic information, Ye et al. [27] designed a margin-based sampling operator to extract attribute features and an attribute-guided decoder to generate corresponding sentences. Besides, Zhang et al. [28] proposed a global visual feature-guided attention mechanism to remove the redundant information in the fused features, followed by LSTM with a linguistic state-guided attention mechanism to generate descriptions.

In comparison to the existing methods, the proposed MVP model in this article introduced an attention-based feature fusion module with the fine-tuned VLP model and the concept of prompt-based learning in RSI captioning. Although prompt-based learning has been introduced in natural image captioning [18], limited attention has been given to bringing prompt-based learning into the RSI domain. A recent study on RSI change captioning achieved this by generating a set of unified prompts and a class-specific prompt to generate captions accordingly for changes in bitemporal RSIs [29]. However, our method has designed a more flexible visual prefix mapping module to generate visual prefixes for the pretrained BERT-based model, followed by generating corresponding captions for RSIs.

III. Method

This section describes the architecture of our method. As shown in Fig. 2, the MVP model is composed of three parts: a MVFF module, a visual prefix mapping module, and a BERT-based caption generator.

A. Multi-View Feature Fusion

Considering that image representations directly affect the quality of the generated caption, we give more attention to two issues: 1) how to extract RSI features and 2) how to enhance feature representations.

For issue 1), we select CLIP [13] and ResNet [30] as the key visual feature extractors to gain two different features. As we mentioned in Section I, both the VP model and the VLP model offer their respective advantages. Specifically, we regard CLIP as our VLP model to capture aligned image-sentence knowledge, while considering ResNet as our VP model to obtain image-based information. To better adapt CLIP to RSI captioning, we first fine-tune the original CLIP model with remote sensing data based on the practice of [13]. During the fine-tuning stage, the original CLIP model is trained on RSI-sentence pairs via contrastive learning, which encourages the model to learn the aligned knowledge between RSIs and sentences. To clarify, we denote the fine-tuned CLIP model as RSCLIP and choose its image encoder to extract the local grid features of RSI, denoted as RSCLIP visual features F_{rsclip}. Additionally, we also use the ResNet-101 model pretrained on the ImageNet dataset to extract the local features of RSI, denoted as ResNet-101 features F_{rn101}. For a given RSI, denoted as I, the process of its visual feature extraction can be formulated as

$$F_{\text{rsclip}} = \text{RSCLIP}_{\text{img}}(I) \tag{1}$$

$$F_{\text{rn101}} = \text{RN101}(I) \tag{2}$$

where $\text{RSCLIP}_{\text{img}}$ represents the RSCLIP image encoder, and RN101 represents the pretrained ResNet-101 model. Besides, $F_{\text{rsclip}} \in \mathbb{R}^{m \times d_1}$, $F_{\text{rn101}} \in \mathbb{R}^{n \times d_2}$, $m \neq n$ and $d_1 \neq d_2$.

To further explore F_{rsclip} and F_{rn101}, we find that they contain rich information about RSI features in different views:

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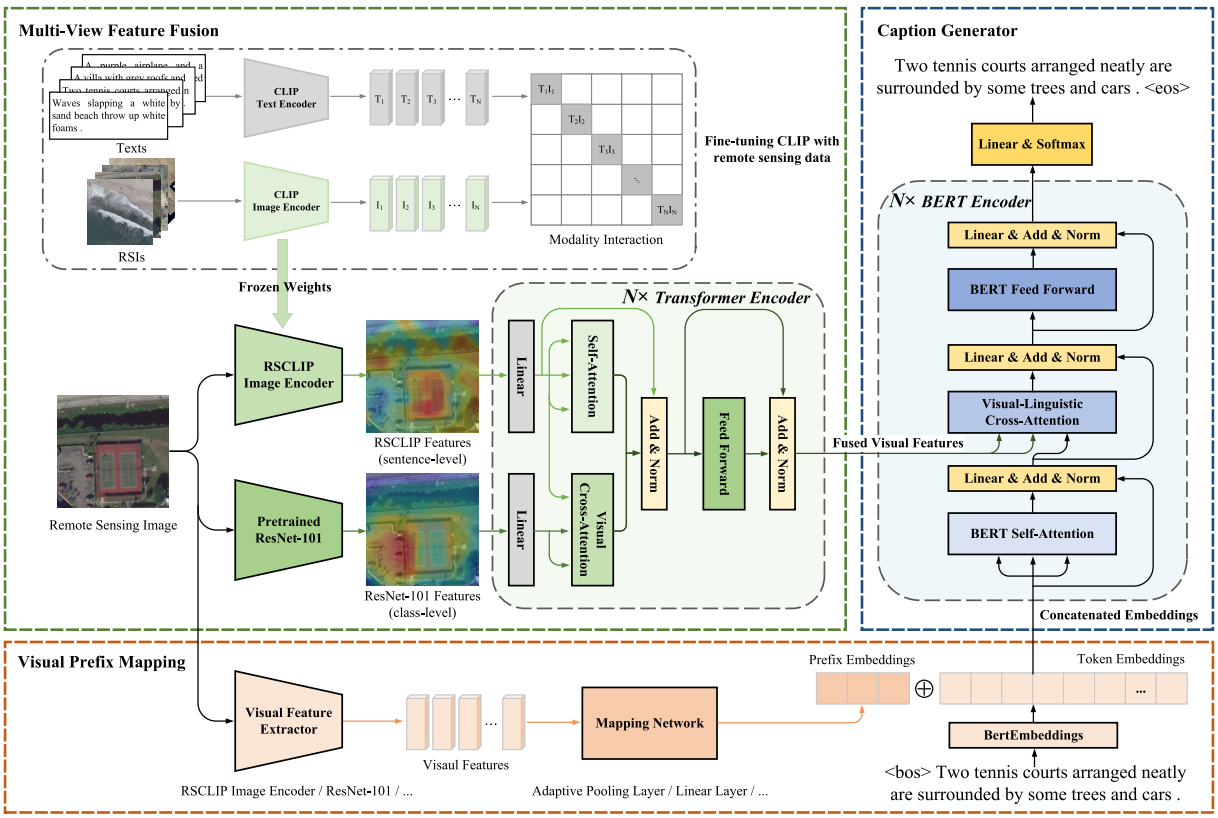


Fig. 2. Architecture of the proposed MVP model. First, a fine-tuned CLIP image encoder and a ResNet-101 network are applied to extract the RSCLIP features and the ResNet-101 features, respectively. Then, two kinds of features are combined as inputs of the attention-based feature fusion module to achieve the fused visual features. Next, we design a visual prefix mapping module to create visual prefix embeddings, which are then concatenated with token embeddings. Finally, both the fused visual features and the concatenated embeddings are fed into BERT encoders to generate captions.

sentence-level and class-level. In traditional terms, the concept of "multi-view" typically refers to considering images from various perspectives, such as different physical sources, positions, or angles. In this article, we interpret "multi-view" to denote multiple representations obtained from various feature extractors describing the same image. Each representation captures a specific aspect of the image. To enhance our understanding of multi-view features, we will provide a detailed introduction to sentence-level features and class-level features as follows. Specifically, the RSCLIP visual features F_{rsclip} are interpreted as sentence-level features, for the reason that they are guided by language supervision and contain the semantic information of RSI-sentence alignment. The ResNet-101 features F_{rn101} are extracted from the pretrained model that is used to be trained with category labels to complete the image classification task. Therefore, the ResNet-101 features F_{rn101} may contain more category information about RSI so that they are interpreted as class-level features. Although we propose the concept of multi-view features to analyze these two different features, we still use F_{rsclip} and F_{rn101} to represent them for clarity.

Regarding issue 2), we propose an attention-based feature fusion module to fuse and enhance multi-view features. First, two linear layers are employed to map F_{rsclip} and F_{rn101} into a d-dimensional space with the corresponding results denoted as $\bar{F}_{\text{rsclip}}$ and $\bar{F}_{\text{rn101}}$, respectively. Then, we select $\bar{F}_{\text{rsclip}}$ as the main features to conduct the multihead self-attention operation [see (3)] and guide the cross-attention operation with $\bar{F}_{\text{rn101}}$

$$\hat{F}_{\text{rsclip}} = \text{MHA}\left(\bar{F}_{\text{rsclip}}, \bar{F}_{\text{rsclip}}, \bar{F}_{\text{rsclip}}\right) \tag{3}$$

$$\hat{F}_{\text{rn101}} = \text{MHA}\left(\bar{F}_{\text{rsclip}}, \bar{F}_{\text{rn101}}, \bar{F}_{\text{rn101}}\right) \tag{4}$$

where MHA$(\cdot)$ represents the multihead attention operation of vanilla transformer, $\hat{F}_{\text{rsclip}} \in \mathbb{R}^{m\times d}$ and $\hat{F}_{\text{rn101}} \in \mathbb{R}^{n\times d}$.

Next, we use add function and layer normalization function to fuse $\bar{F}_{\text{rsclip}}$, $\hat{F}_{\text{rsclip}}$ and $\hat{F}_{\text{rn101}}$ as follows:

$$F_{\text{im}} = \text{LayerNorm}\left(\bar{F}_{\text{rsclip}} + \hat{F}_{\text{rsclip}} + \hat{F}_{\text{rn101}}\right). \tag{5}$$

Finally, the intermediate fused features F_{im} are fed into the FeedForward module to generate the final fused visual features F_{fusion}, which subsequently interact with the caption generator module. It is worth mentioning that the attention-based feature fusion module, a variant of the transformer encoder, is a N-layer stacked deep network. We attempt to apply these deep networks for better feature fusion and explore the potential information of the fused visual features.

B. Visual Prefix Mapping

To produce visual prefix embeddings, we propose a flexible two-submodule framework, called visual prefix mapping. Inspired by the ideas of prompt-based learning, we desire to make full use of RSIs and create the visual prefixes, which

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are regarded as visual prompt information for subsequent sentence generation. Therefore, a two-submodule framework is designed to fulfill the goal of producing visual prefixes. As shown at the bottom of Fig. 2, our two-submodule framework consists of a visual feature extractor and a mapping network. Besides, the flexibility of our framework can be further explained as follows.

To begin with, we use a visual feature extractor, denoted as G_{vis}, to gain visual information of a given RSI. Any network that has the ability to transform an RSI into feature vectors can be considered a visual feature extractor. As a result, a general formula is used to represent the operation of the visual feature extractor

$$F_{\text{mid}} = G_{\text{vis}}(I) \tag{6}$$

where $F_{\text{mid}} \in \mathbb{R}^{m \times d_m}$ refers to the visual features extracted from the visual feature extractor.

Then, we adopt a mapping network G_{map} to map the visual features to k embedding vectors, called visual prefix embeddings P_k. Likewise, any network or operation that is able to transform visual features into the specified dimension vectors can be considered a mapping network. The mapping process can be expressed as

$$\text{P}_k = p_1, p_2, \ldots, p_k = G_{\text{map}}(F_{\text{mid}}). \tag{7}$$

Noted that each embedding vector p_i has the same d_w dimension as a word embedding and $\text{P}_k \in \mathbb{R}^{k \times d_w}$.

Moreover, we employ the BertEmbeddings module [31] to gain textual information of raw sentences during training. The output vectors are described as token embeddings T_l [see (8)], where l is the length of tokens

$$T_l = t_1, t_2, \ldots, t_l. \tag{8}$$

Finally, we concatenate the obtained visual prefix embeddings P_k with the token embeddings T_l [see (9)]. The prefix-token concatenated embeddings E_{cat} are then fed into the caption generator

$$E_{\text{cat}} = p_1, p_2, \ldots, p_k, \quad t_1, t_2, \ldots, t_l. \tag{9}$$

To conclude, we propose a flexible framework consisting of two optional submodules to obtain visual prefixes. Due to the diverse choices of two submodules, visual prefix embeddings produced by various combined networks may have different effects on caption generation. To further understand our proposed framework in this part, we give some combination examples in Fig. 3.

In this article, we select the RSCLIP image encoder as the visual feature extractor and choose adaptive average pooling layers as the mapping network to finally generate visual prefix embeddings. The reason why we pick the RSCLIP image encoder is that the extracted RSCLIP visual features contain more sentence-level information, which may guide the predicted sentence generation. Besides, unlike common pooling operations, adaptive average pooling operation accepts any given output size for average downsampling. Accordingly, adaptive average pooling layers can better map the extracted visual features into the word space in terms of feature embedding. It is worth mentioning that different combined networks would be chosen when facing different situations (e.g., natural image captioning, non-English image captioning). The effectiveness of the combination will be further demonstrated through the ablation study in Section IV-E2.

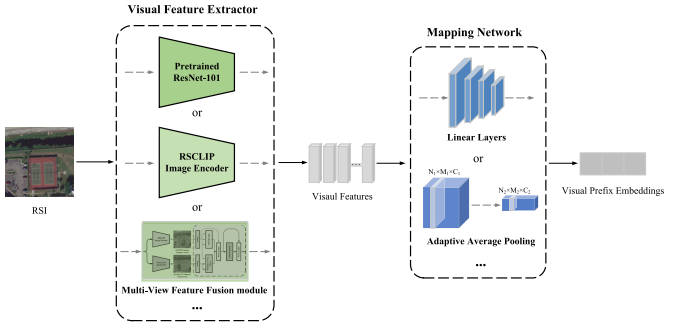


Fig. 3. Framework of the designed visual prefix mapping module. It consists of a visual feature extractor and a mapping network, which can extract the visual features from RSI and transform them into the specified dimension vector. Some examples are presented in the figure, showing that the pretrained ResNet-101 model, the RSCLIP image encoder, or the MVFF module can be selected as the visual feature extractor, while linear layers or adaptive average pooling operations can be chosen as the mapping network.

C. BERT-Based Caption Generator

The aforementioned explanations have illustrated how we obtain the fused visual features F_{fusion} and the concatenated embeddings E_{cat}. For the fused visual features F_{fusion}, we first use the RSCLIP image encoder and the pretrained ResNet-101 model to extract the features of RSIs, respectively. Then, an attention-based feature fusion module is employed to fuse these two features, producing the final fused visual features. For the concatenated embeddings E_{cat}, we adopt a two-submodule framework to produce the visual prefix embeddings, which are concatenated with token embeddings to constitute the final prefix-token concatenated embeddings. However, how to generate high-quality captions using these two important pieces of information is what we concentrate on. Thus, we turn to the language model BERT [31], which is good at text processing.

In this part, we propose a BERT-based caption generator to combine the fused visual features F_{fusion} and the concatenated embeddings E_{cat} to predict the corresponding captions of RSIs. As shown at the top-right of Fig. 2, the concatenated embeddings E_{cat} are first fed into the BERT encoders to conduct the multihead self-attention operation, followed by the output operation: a linear layer, the add function, and layer normalization function

$$H_{\text{seq}} = \text{LayerNorm}((\text{Linear}(\text{MHA}(E_{\text{cat}})) + E_{\text{cat}})) \tag{10}$$

where H_{seq} refers to the output of the multihead self-attention operation $\text{MHA}(\cdot)$ with the general output operations.

Then, H_{seq}, which acts as a query, is passed to guide the visual-linguistic cross-attention operation with the fused visual features F_{fusion}

$$H_{\text{vl}} = \text{MHA}\big(H_{\text{seq}}, F_{\text{fusion}}, F_{\text{fusion}}\big). \tag{11}$$

Subsequently, the intermediate fused visual-linguistic features H_{vl} are fed into the BERT FeedForward module, which

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is the last part of the BERT encoder. We denote the number of the BERT encoders as N and the output of the Nth BERT encoder as H_N. Finally, the process of word prediction can be formulated as

$$p(w_t|p_{1:k}, w_{1:t-1}) = \text{Softmax}\left(W^D H_N\right) \tag{12}$$

$$H_t = \text{BERT}_{\text{Enc}}(F_{\text{fusion}}, H_{t-1}), \quad H_1 = E_{\text{cat}} \tag{13}$$

where $p_{1:k}$ denotes the visual prefix embeddings and $p(w_t|p_{1:k}, w_{1:t-1})$ represents the probability of the tth predicted word conditioned on the visual prefix embeddings. BERT_{Enc} denotes all operations in a BERT encoder and h_t refers to the output of the tth BERT encoder. $W^D \in \mathbb{R}^{V\times d_h}$ is trainable parameters, in which V is the vocabulary size and d_h is the hidden size of the BERT encoder.

In addition, to avoid significant changes between training and inference, we modify the standard BERT encoder with a special attention mask to predict the next token one by one. Specifically, the attention mask operation ensures that the prediction of each token depends on the preceding generated tokens while having no access to the subsequent tokens or the padding tokens.

D. Training Objective and Inference Details

During training, all layers in both the attention-based feature fusion module and the BERT-based caption generator are trained. We introduce the typical cross-entropy loss L_{CE} to optimize our model

$$L_{\text{CE}}(\theta) = -\sum_{t=1}^{T} \log\left(p_\theta\left(y_t^*|y_{1:t-1}^*\right)\right) \tag{14}$$

where $y_{1:t-1}^*$ denotes the target ground-truth sequence, T is the length of the sentence, and θ represents the model parameters.

Since our training objective focuses on sentence optimization without feature extractor optimization, the trained RSCLIP image encoder and the pretrained ResNet-101 model are not involved in the training process but are directly employed for image feature extraction. To avoid redundant feature extraction during both training and inference, we store the extracted features locally and load them before performing MVFF. This operation accelerates both training and inference.

During inference, the multi-view features extracted from the RSCLIP image encoder and the pretrained ResNet-101 model are fed into the trained attention-based feature fusion module to dynamically produce the fused visual features. Besides, the visual prefix embeddings P_k of the RSI are obtained using the selected visual feature extractor G_{vis} and the mapping network G_{map}. Our model generates sentences conditioned on the visual prefix embeddings and predicts the next tokens one by one under the guidance of the trained BERT-based caption generator. Furthermore, the beam search method is also applied to boost the diversity of generated captions to some extent.

As shown in Fig. 4, it is important to emphasize that the training and inference process of our method is consistent, and there are no separate modules specifically designed for training or inference.

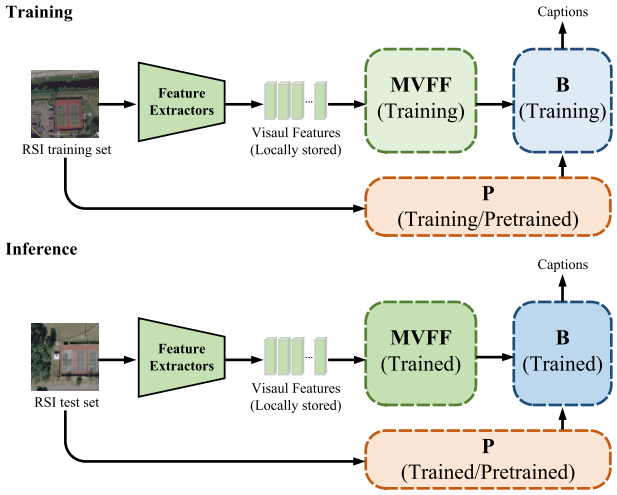


Fig. 4. Training and inference process of our model. "MVFF," "P," and "B" refer to the MVFF module, the visual prefix mapping module, and the Bert-based caption generator, respectively. The contents in the brackets represent the status of the current modules. Due to various combinations in the visual prefix mapping modules, the status may be "Training" or "Pretrained" in the training stage.

IV. Experiments

A. Datasets

To evaluate the effectiveness of our proposed method, we conduct extensive experiments on three well-known public datasets: the UCM-Captions dataset [1], the Sydney-Captions dataset [1], and the RSICD [2]. Particularly, the SEN1-2 dataset [32] is not involved in the quantitative experiments but will be used in the subsequent qualitative experiments.

1) UCM-Captions: Based on the UC Merced Land-Use data [33], the UCM-Captions dataset contains 2100 high-resolution images divided into 21 land-use classes. Each class is assigned 100 images and all images measure 256 × 256 pixels with a pixel resolution of a foot. Then, every image is annotated with five corresponding descriptions.

2) Sydney-Captions: Based on the Sydney dataset [34], the Sydney-Caption dataset contains 613 images acquired from Google Earth of Sydney, Australia. Each image is cropped to a size of 500 × 500 pixels with a pixel resolution of 0.5 m. There are five different reference descriptions for every image.

3) RSICD: This RSI dataset is by far the largest, with all the images collected from Google Earth, Baidu Map, MapABC, and Tianditu. It contains 10 921 images, which are fixed to 224 × 224 pixels with various resolutions. To ensure every image has five descriptions, Lu et al. [2] extend the sentences by randomly duplicating the existing sentences. Because of its large and diverse type, this dataset supports the researchers to further advance RSI captioning.

4) SEN1-2: This dataset contains 282 384 pairs of corresponding SAR and optical images acquired by the Sentinel-1 and Sentinel-2 satellites, respectively. These images are collected from across the globe and throughout all meteorological seasons. To further explore the method in real-world scenarios, optical images from the Sentinel-2 satellite, each measuring 256 × 256 pixels with a pixel resolution of 10 m, will be used in the qualitative analysis.

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B. Evaluation Metrics

Several evaluation metrics are used in this article, including BLEU-n [35], METEOR [36], ROUGE_L [37], CIDEr [38], SPICE [39], and S_m. Each metric has a different focus on the caption quality, while the S_m metric represents the average performance of caption models. The higher metric scores mean better caption model performance.

1) BLEU-n: The bilingual evaluation understudy (BLEU) metric is first applied to measure machine translation performance. It focuses on the co-occurrences between the predicted caption and the ground-truth sentence by using n-grams matching (usually $n = 1, 2, 3, 4$).

2) METEOR: The metric for evaluation of translation with explicit ordering (METEOR) is also an automatic metric for machine translation evaluation. Unlike n-grams-based BLEU, METEOR is designed on a generalized concept of unigram matching, such as the matching of word surface forms, word stemmed forms, and synonyms, between the predicted caption and the ground-truth sentence. The longer the n-grams, the fewer chunks, which refers to a series of contiguous order matches.

3) ROUGE_L: The recall-oriented understudy for gisting evaluation (ROUGE) metric is used to measure the similarity between summaries. Specifically, ROUGE_L, one of the ROUGE measures, is calculated by an F-measure with a recall bias using the longest common subsequence (LCS) between the predicted caption and the ground-truth sentence.

4) CIDEr: The consensus-based image description evaluation (CIDEr) metric is specially designed for image captioning tasks. It measures the consensus by using a term frequency-inverse document frequency (TF-IDF) weighting for each n-gram between the predicted caption and the ground-truth sentence.

5) SPICE: The semantic propositional image caption evaluation (SPICE) metric is also a principled metric for automatic image caption evaluation, which focuses on the semantic contents, such as objects, attributes, and relationships in the sentences. This metric is calculated using an F-score of the matching tuples in the predicted caption and ground-truth sentence scene graphs.

6) S_m: This metric was first proposed in the 2017 AI Challenger,[1] standing for an overall evaluation metric of the image caption model performance. It is calculated by the mean of BLEU-4, METEOR, ROUGE, and CIDEr as follows:

$$S_m = \frac{1}{4}(\text{BLEU} - 4 + \text{METEOR} + \text{ROUGE_}L + \text{CIDEr}). \tag{15}$$

Additionally, we would like to clarify that the listed metrics in our paper vary in their level of importance. CIDEr and SPICE metrics are designed specifically for image captioning, while others, such as BLEU and METEOR, are adapted from machine translation, and ROUGE_L is adapted from text summarization. S_m is a comprehensive evaluation metric. Therefore, CIDEr and SPICE scores are more reliable for image captioning, while S_m scores provide insight into the overall performance of the model. We will particularly emphasize the scores of these three metrics in the subsequent experiment analysis.

[1]https://challenger.ai/competition/caption

C. Implementation Details

For each dataset, we divide into three subsets: 80% of the image–sentence pairs are used as the training set, 10% as the validation set, and 10% as the test set. These divisions are consistent with the compared methods to ensure a fair comparison. All the images are resized to 224 × 224 pixels in image preprocessing, while a new vocabulary is constructed by selecting the words that appear more than five times in sentence preprocessing.

In preparation for the feature extraction, we choose Transformer and ViT-B/16 as the text encoder and the image encoder of the original CLIP model, respectively. ViT-B/16 means the base version of Visual Transformer [23] with 16 × 16 input patch size. During the fine-tuning stage, the original CLIP model is trained on RSI–sentence pairs using a contrastive loss function. The resulting best model from fine-tuning is named RSCLIP. It is important to emphasize that RSCLIP is trained separately on three datasets, meaning each dataset has its dedicated RSCLIP. For each dataset, RSCLIP is exclusively trained on the training set and evaluated on the validation set using retrieval metrics. This operation ensures consistent dataset divisions in our experiments and prevents access to the test set. To represent the RSI features, we use the RSCLIP image encoder to extract the 768-D grid features of RSI while using the pretrained ResNet-101 model to extract the 2048-D local features of RSI from the convolution layer. To further explain the features extracted from the pretrained ResNet-101 model, we initially extract them from the last convolutional layer, resulting in the size of 14 × 14 × 2048, and subsequently reshape them to the size of $n \times d_2$, specifically 196 × 2048. Then, two linear layers with a dimension of 512 are employed to map both extracted features into identical dimensional vectors. The number of layers in the MVFF module and the number of BERT encoders are both set to six ($N = 6$). Besides, we use eight heads in each multihead attention operation and assign the dimension of the FeedForward layer to 2048. For the visual prefix embeddings $\mathrm{P}_k \in \mathbb{R}^{k \times d_w}$, we set the length of embeddings to $k = 5$ and the dimension to $d_w = 768$.

In the training stage, we train our model with a batch size of 32 for 35 epochs. Upon careful analysis, we find that a small input batch size can lead to erratic convergence and potentially unstable gradient estimates. Therefore, a batch size of 32 has been identified as a common and effective choice across all datasets in our experiments. The RAdam optimizer [40] is selected to update the model parameters with an initial learning rate of 5e-4. In the inference stage, the beam size of 5 is applied. Our experiments are implemented on the Nvidia GeForce RTX 3090 with PyTorch version 1.8.0.

D. Comparison With State-of-the-Art Methods

We compare our proposed method with several SOTA methods, including soft attention [2], hard attention [2], FC-Att + LSTM [19], SM-Att + LSTM [19], SAT (LAM)* [20],

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TABLE I
COMPARISON RESULTS WITH STATE-OF-THE-ART METHODS ON THE UCM-CAPTIONS DATASET

Method	Enc-Dec	*B*1	*B*2	*B*3	*B*4	*M*	*R*	*C*	*S*	S_m
Soft Attention [2]	VGG-16+LSTM	74.54	65.54	58.55	52.5	38.86	72.37	261.24	-	106.24
Hard Attention [2]	VGG-16+LSTM	81.57	73.12	67.02	61.82	42.63	76.98	299.47	-	120.23
FC-Att+LSTM [19]	VGG-16+LSTM	81.35	75.02	68.49	63.52	41.73	75.04	299.58	-	119.97
SM-Att+LSTM [19]	VGG-16+LSTM	81.54	75.75	69.36	64.58	42.40	76.32	318.64	-	125.49
SAT (LAM)* [20]	VGG-16+LSTM	81.95	77.64	74.85	71.61	48.37	79.08	361.71	50.24	140.19
Adaptive (LAM)* [20]	VGG-16+LSTM	81.70	75.10	69.90	65.40	44.80	78.70	328.00	50.30	129.23
Structured Attention [8]	ResNet-50+LSTM	85.38	80.35	75.72	71.49	46.32	81.41	334.89	-	133.53
MLA [7]	ResNet-101+LSTM	84.06	78.03	73.33	69.16	**53.30**	81.96	311.93	46.01	129.09
RASG [6]	VGG-16+LSTM	85.18	79.25	74.32	69.76	45.71	80.72	338.87	48.91	133.77
VRTMM [9]	VGG-16+Transformer	83.94	77.85	72.83	68.28	45.27	80.26	349.48	-	135.82
GVFGA+LSGA [28]	CNN+LSTM	83.19	76.57	71.03	65.96	44.36	78.45	332.70	48.53	130.37
GLCM [41]	GoogLeNet+Transformer	81.82	75.40	69.86	64.68	46.19	75.24	302.79	-	122.23
DFEN [42]	ResNet-18+LSTM	85.10	78.40	72.80	67.70	45.90	80.50	317.70	50.10	127.95
MSISAM [43]	VGG-16+LSTM	87.27	80.96	75.51	70.39	46.52	82.58	371.29	-	142.70
MVP	Transformer+BERT	**89.05**	**84.21**	**79.90**	**75.89**	48.97	**84.41**	**379.05**	**52.78**	**147.08**
Δ		+1.78	+3.25	+4.18	+4.28	-4.33	+1.83	+7.76	+2.48	+4.38

"-" means the SPICE scores are not reported in the reference papers, so the value of *S* will be omitted correspondingly.
"Δ" means the improvement of our method compared to the best scores achieved by other methods.

Adaptive (LAM)* [20], structured attention [8], MLA [7], RASG [6], VRTMM [9], GLCM [41], GVFGA + LSGA [28], DFEN [42], and MSISAM [43]. These SOTA methods are all based on an encoder–decoder architecture with different deep networks and attention mechanisms. We give a brief introduction of them as follows.

Soft attention and hard attention are both attention-based methods using VGG-16 + LSTM architecture. The "soft" attention mechanism gives weights to different parts of an RSI according to the former hidden state, while the "hard" attention mechanism adopts a sampling strategy to consider different parts of an RSI, followed by using reinforcement learning. Both two attention mechanisms are proposed in [44].

FC-Att + LSTM and SM-Att + LSTM are two variants of the attribute attention-based method. FC-Att + LSTM uses the output of the last fully connected layer of VGG-16 as attributes to perform the attention mechanism, while SM-Att + LSTM uses the output of the softmax layer of VGG-16 as attributes.

SAT (LAM)* and Adaptive (LAM)* both adopt a label-attention mechanism, which can utilize label information from the given image to calculate the attention masks and encourage the attended image features more salient. "SAT" stands for the "Show-Attend-and-Tell" model proposed in [44], while "Adaptive" stands for the model proposed in [45]. It is noted that the preprocessing of experimental datasets is different from other compared methods, so we use "*" to specifically mark its difference.

Structured attention is a method using ResNet-50 + LSTM architecture with the structured attention mechanism. The structured attention module is aimed at utilizing the structured information of semantic contents in RSIs and interacting with the decoder during caption generation.

MLA refers to a multilevel attention-based model. The model contains three attention structures to perform multilevel attention, which automatically decides to focus on the image or the language. Besides, the pretrained ResNet-101 model is used as the encoder to extract the convolution layer features of the image and select LSTM as the decoder.

RASG represents the given RSI by concatenating original CNN features and the global feature of the multiscale features. Based on a two-layer LSTM, the decoder in RASG integrates a recurrent attention mechanism and a semantic gate to generate a better context vector.

VRTMM is a two-stage model based on the variational autoencoder and reinforcement learning. Image features are extracted from the VGG-16, which is fine-tuned with a variational autoencoder. Instead of LSTM, Transformer is employed as the decoder to generate descriptions.

GVFGA + LSGA refers to a CNN + LSTM architecture that utilizes a global visual feature-guided attention mechanism and a linguistic state-guided attention mechanism. These mechanisms are proposed to eliminate redundant information from fused features and refine the fusion of visual and textual features, respectively.

GLCM is an attention-based global–local captioning model that acquires global–local visual feature representation for RSI captioning, aiming to leverage the benefits of both global and local features. Additionally, GLCM employs a visual-linguistic attention mechanism with the corresponding attention-based visualization.

DFEN is a dual feature enhancement network that employs image-enhance and text-enhance modules to capture ground object information from both image and text perspectives. The image-enhance module is applied after extracting global features from the pretrained ResNet-18 model, while the text-enhance module, along with a two-layer LSTM, is designed to generate corresponding captions of RSIs.

MSISAM stands for a multisource interactive stair attention network, which independently models the semantics of preceding sentences and visual regions of interest. The multisource interactive attention network can selectively attend to the feature maps from CNNs, while the stair attention network is designed to redistribute the existing attention weights.

1) Results on the UCM-Captions Dataset: Table I presents the comparison results of our proposed method and other SOTA methods on the UCM-Captions dataset. Except for

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TABLE II
COMPARISON RESULTS WITH STATE-OF-THE-ART METHODS ON THE SYDNEY-CAPTIONS DATASET

Method	Enc-Dec	*B1*	*B2*	*B3*	*B4*	*M*	*R*	*C*	*S*	S_m
Soft Attention [2]	VGG-16+LSTM	73.22	66.74	62.23	58.20	39.42	71.27	249.93	-	104.71
Hard Attention [2]	VGG-16+LSTM	75.91	66.10	58.89	52.58	38.98	71.89	218.19	-	95.41
FC-Att+LSTM [19]	VGG-16+LSTM	80.76	71.60	62.76	55.44	40.99	71.14	220.33	-	96.98
SM-Att+LSTM [19]	VGG-16+LSTM	81.43	73.51	65.86	58.06	41.11	71.95	230.21	-	100.33
SAT (LAM)* [20]	VGG-16+LSTM	74.05	65.50	59.04	53.04	36.89	68.14	235.19	40.38	98.32
Adaptive (LAM)* [20]	VGG-16+LSTM	73.23	63.16	56.29	50.74	36.13	67.75	234.55	42.43	97.29
Structured Attention [8]	ResNet-50+LSTM	77.95	70.19	63.92	58.61	39.54	72.99	237.91	-	102.26
MLA [7]	ResNet-101+LSTM	81.52	**74.44**	67.55	61.39	**45.60**	70.62	199.24	38.90	94.21
RASG [6]	VGG-16+LSTM	80.00	72.17	65.31	59.09	39.08	72.18	263.11	43.01	108.37
VRTMM [9]	VGG-16+Transformer	74.43	67.23	61.72	56.99	37.48	66.98	252.85	-	103.58
GVFGA+LSGA [28]	CNN+LSTM	76.81	68.46	61.45	55.04	38.66	70.30	245.22	45.32	102.31
GLCM [41]	GoogLeNet+Transformer	80.41	73.05	67.45	**62.59**	44.21	69.65	243.37	-	104.96
DFEN [42]	ResNet-18+LSTM	79.80	69.70	61.40	54.20	37.30	72.30	200.90	44.90	91.18
MSISAM [43]	VGG-16+LSTM	76.43	69.19	62.83	57.25	39.46	71.72	**281.22**	**-**	**112.41**
MVP	Transformer+BERT	**81.54**	74.32	**67.93**	61.82	41.07	**74.45**	260.49	**48.33**	109.46
Δ		+0.02	-0.12	+0.38	-0.77	-4.53	+1.46	-20.73	+3.01	-2.95

"-" means the SPICE scores are not reported in the reference papers, so the value of *S* will be omitted correspondingly.
"Δ" means the improvement of our method compared to the best scores achieved by other methods.

TABLE III
COMPARISON RESULTS WITH STATE-OF-THE-ART METHODS ON THE RSICD

Method	Enc-Dec	*B1*	*B2*	*B3*	*B4*	*M*	*R*	*C*	*S*	S_m
Soft Attention [2]	VGG-16+LSTM	67.53	53.08	43.33	36.17	32.55	61.09	196.43	-	81.56
Hard Attention [2]	VGG-16+LSTM	66.69	51.82	41.64	34.07	32.01	60.84	179.25	-	76.54
FC-Att+LSTM [19]	VGG-16+LSTM	74.59	62.50	53.38	45.74	33.95	63.33	236.64	-	94.92
SM-Att+LSTM [19]	VGG-16+LSTM	75.71	63.36	53.85	46.12	35.13	64.58	235.63	-	95.37
SAT (LAM)* [20]	VGG-16+LSTM	67.53	55.37	46.86	40.26	32.54	58.23	258.50	46.36	97.38
Adaptive (LAM)* [20]	VGG-16+LSTM	66.64	54.86	46.76	40.70	32.30	58.43	260.55	46.73	98.00
Structured Attention [8]	ResNet-50+LSTM	70.16	56.14	46.48	39.34	32.91	57.06	170.31	-	74.91
MLA [7]	ResNet-101+LSTM	77.25	62.90	53.28	46.08	34.71	69.10	236.37	43.15	96.57
RASG [6]	VGG-16+LSTM	77.29	66.51	57.82	50.62	36.26	66.91	275.49	47.19	107.32
VRTMM [9]	VGG-16+Transformer	78.13	67.21	56.45	51.23	**37.37**	67.13	271.50	-	106.81
GVFGA+LSGA [28]	CNN+LSTM	67.79	56.00	47.81	41.65	32.85	59.29	260.12	46.83	98.48
GLCM [41]	GoogLeNet+Transformer	77.67	64.92	56.42	49.37	36.27	67.79	254.91	-	102.09
DFEN [42]	ResNet-18+LSTM	76.60	63.60	53.80	46.30	37.30	68.50	260.50	47.70	103.15
MSISAM [43]	VGG-16+LSTM	78.36	66.79	57.74	50.42	36.72	67.30	284.36	-	109.70
MVP	Transformer+BERT	**79.80**	**69.56**	**61.00**	**53.69**	37.34	**69.21**	**292.48**	**49.31**	**113.18**
Δ		+1.44	+2.35	+3.18	+2.46	-0.03	+0.11	+8.12	+1.61	+3.48

"-" means the SPICE scores are not reported in the reference papers, so the value of *S* will be omitted correspondingly.
"Δ" means the improvement of our method compared to the best scores achieved by other methods.

METEOR, our method outperforms the other methods on all metrics by a significant margin. Specifically, it is observed that our method exceeds the previous methods by 5.98% on BLEU-4, 2.22% on ROUGE_L, and 4.93% on SPICE. What stands out in this table is the CIDEr score, which is a 379.05 score achieved by our method and is 2.09% higher than the MSISAM method. Although the METEOR score of our method performs poorer than that of the best method, MLA, it wins the second place with a score of 48.97 points. For the S_m score, our method reaches the best score in comparison with other SOTA methods, which demonstrates that the MVP model has a better overall performance to some extent.

2) Results on the Sydney-Captions Dataset: As shown in Table II, our method ranks first in four-ninths of the evaluation metrics on the Sydney-Captions dataset. More specifically, our method achieves competitive scores on BLEU-1, BLEU-3, and ROUGE_L in comparison with those of the best methods, while being slightly worse on BLEU-2, BLEU-4, and METEOR. The reason for our model's lower performance in terms of the mentioned three metrics may be attributed to the susceptibility of the Transformer-based model to overfitting on small-size datasets. Although our method achieves suboptimal results for the mentioned three metrics, it significantly outperforms other methods in terms of the SPICE metrics. Particularly, the SPICE score improvement achieved by our method is 3.01 points, indicating that our method beats other SOTA methods in aspects of describing semantic contents to some degree. For the S_m score, the proposed method secures second place with a score of 109.46, demonstrating the overall superiority of our method.

3) Results on the RSICD: In Table III, it is seen that our proposed method surpasses other SOTA methods significantly in terms of nearly all metrics on the RSICD. In terms of the CIDEr score, our method achieves a score of 292.48, demonstrating a 2.86% improvement over the best method, MSISAM. While our method is not top-ranked in terms of METEOR, it closely follows the best-performing method, VRTMM, with only a 0.03-point difference. After careful

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TABLE IV
ABLATION RESULTS USING DIFFERENT CLASS-LEVEL FEATURES ON THREE DATASETS

Dataset	Model	*B*1	*B*2	*B*3	*B*4	*M*	*R*	*C*	*S*	S_m
UCM-Captions	Transformer+VGG16	72.33	64.03	57.90	52.55	38.11	70.53	265.41	40.43	106.65
	Transformer+VGG19	76.55	70.02	64.43	59.03	40.11	72.16	274.24	41.40	111.39
	Transformer+DenseNet121	77.63	71.05	**66.11**	**61.90**	38.88	72.28	281.97	40.62	113.76
	Transformer+DenseNet169	74.69	66.51	60.39	55.39	39.90	71.56	281.04	**42.39**	111.97
	Transformer+RN101	**79.80**	**72.44**	66.05	60.19	**40.44**	**75.14**	**288.65**	40.50	**116.11**
Sydney-Captions	Transformer+VGG16	66.15	55.70	48.71	43.73	32.70	59.48	184.97	35.58	80.22
	Transformer+VGG19	69.10	58.24	50.78	44.84	33.23	**62.29**	185.51	40.10	81.47
	Transformer+DenseNet121	68.55	57.87	50.42	44.04	33.22	61.05	175.57	36.36	78.47
	Transformer+DenseNet169	68.94	59.40	52.12	45.93	**34.14**	61.94	176.34	**40.61**	79.59
	Transformer+RN101	**70.21**	**60.52**	**53.08**	**47.03**	32.24	60.01	**193.62**	39.71	**83.23**
RSICD	Transformer+VGG16	73.83	62.71	54.17	47.24	35.98	65.52	252.46	46.12	100.30
	Transformer+VGG19	73.32	61.68	52.80	45.56	35.72	64.91	246.86	45.20	98.26
	Transformer+DenseNet121	**77.32**	**65.80**	**56.41**	48.71	36.59	66.31	262.84	47.51	103.61
	Transformer+DenseNet169	76.44	64.70	55.62	48.09	36.24	65.76	257.52	46.79	101.90
	Transformer+RN101	76.45	65.23	56.39	**48.96**	**36.79**	**66.39**	**267.10**	**47.68**	**104.81**

TABLE V
ABLATION RESULTS USING VISUAL FEATURES EXTRACTED FROM DIFFERENT CLIP IMAGE ENCODERS ON THREE DATASETS

Dataset	Model	*B*1	*B*2	*B*3	*B*4	*M*	*R*	*C*	*S*	S_m
UCM-Captions	Transformer+CLIP(RN50)	78.69	72.19	66.84	62.35	40.18	74.77	309.33	44.55	121.66
	Transformer+CLIP(RN101)	80.55	74.05	68.47	63.43	**43.17**	**77.60**	315.15	45.59	124.84
	Transformer+CLIP(ViT-B/16)	**83.03**	**76.60**	**70.90**	**65.64**	42.92	77.59	**320.24**	**47.32**	**126.60**
Sydney-Captions	Transformer+CLIP(RN50)	69.13	57.95	49.95	43.18	32.68	60.30	181.77	38.96	79.48
	Transformer+CLIP(RN101)	**72.71**	63.13	**56.55**	**51.71**	36.05	65.04	223.67	41.04	94.12
	Transformer+CLIP(ViT-B/16)	72.44	**63.69**	56.41	50.09	**36.54**	**68.06**	**239.53**	**44.56**	**98.56**
RSICD	Transformer+CLIP(RN50)	77.22	66.23	57.57	50.38	38.36	69.12	281.57	49.60	109.86
	Transformer+CLIP(RN101)	79.75	**68.79**	**60.13**	**52.95**	38.57	**69.84**	287.78	**50.18**	112.29
	Transformer+CLIP(ViT-B/16)	**79.83**	68.76	59.75	52.20	**38.80**	69.44	**289.07**	49.98	**112.38**

analysis, the lower METEOR metric performance of our model may be due to the increased number of chunks in the generated sentences, resulting in decreased METEOR scores. Although our method provides accurate and comprehensive descriptions, it is observed that the sentences it generates have fewer matched unigrams compared to the ground-truth sentences, resulting in an increased number of chunks and decreased METEOR scores. For the S_m score, our method outperforms the previous best method by a significant margin of 3.48 points, demonstrating the overall superiority of the proposed method.

E. Ablation Studies

In this part, we analyze the advantage of our MVP model in detail to demonstrate the effectiveness of the fused visual features, the visual prefixes, and different designed modules.

1) Visual Features Discussion: To explore the performance of different visual features, we design the ablation experiments and analyze the results in this part. All the ablation experiments are based on identical Transformer settings, where the number of attention blocks is set to 6, on three datasets. We further illustrate the experiments and models as follows.

Table IV displays ablation experiments on different class-level features. The experiments use identical Transformer settings to compare visual features extracted from ResNet-101, VGG-16 [46], VGG-19 [46], DenseNet-121 [47], and DenseNet-169 [47] on three datasets. It is worth mentioning that **"Transformer + *A*"** in this article denotes the original Transformer model with the visual features of RSIs extracted from the *A* model. For example, **"Transformer + RN101"** represents the Transformer model using the ResNet-101 features of RSIs extracted from the pretrained ResNet-101 model. It is observed that "Transformer + RN101" surpasses other models on most metric scores, especially on the CIDEr score and the S_m score on three datasets. Although several scores in "Transformer + RN101" do not rank first, like BLEU-3, BLEU-4, and SPICE, in the UCM-Captions dataset, they are slightly worse than the best scores with a gap of less than 1.90 points. The overall results indicate that the ResNet-101 features may have better class-level performance in comparison with other visual features and we select them as the class-level features in the experiments.

Additionally, we conduct ablation experiments on visual features extracted from various CLIP image encoders, given the significance of CLIP features. There are three types of CLIP image encoders used in the experiments: ResNet-50, ResNet-101, and ViT-B/16. In Table V, we observe that "Transformer + CLIP(ViT-B/16)" outperforms "Transformer + CLIP(RN50)" and "Transformer + CLIP(RN101)" on the UCM-Captions and Sydney-Captions datasets, while it performs competitively with "Transformer + CLIP(RN101)" on the RSICD. Given the superior performance of the CLIP features extracted from ViT-B/16, we choose ViT-B/16 as the

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TABLE VI
ABLATION RESULTS WITH DIFFERENT VISUAL FEATURES ON THREE DATASETS

Dataset	Model	*B*1	*B*2	*B*3	*B*4	*M*	*R*	*C*	*S*	S_m
UCM-Captions	Transformer+RN101	79.80	72.44	66.05	60.19	40.44	75.14	288.65	40.50	116.11
	Transformer+CLIP	83.03	76.60	70.90	65.64	42.92	77.59	320.24	47.32	126.60
	Transformer+RSCLIP	**88.16**	**82.35**	77.30	72.70	46.64	82.31	349.04	49.50	137.67
	Transformer+CLIP+RN101	81.86	75.28	69.64	64.61	45.34	79.40	332.12	48.78	130.37
	Transformer+RSCLIP+RN101	88.05	82.23	**77.65**	**73.58**	**47.94**	**82.79**	**362.59**	**51.94**	**141.73**
Sydney-Captions	Transformer+RN101	70.21	60.52	53.08	47.03	32.24	60.01	193.62	39.71	83.23
	Transformer+CLIP	72.44	63.69	56.41	50.09	36.54	68.06	239.53	44.56	98.56
	Transformer+RSCLIP	78.25	69.04	61.30	54.53	38.46	69.85	245.13	44.59	101.99
	Transformer+CLIP+RN101	74.60	65.26	57.61	51.01	36.16	67.74	237.55	44.80	98.12
	Transformer+RSCLIP+RN101	**79.55**	**72.21**	**65.68**	**59.94**	**40.45**	**73.26**	**245.40**	**45.97**	**104.76**
RSICD	Transformer+RN101	76.45	65.23	56.39	48.96	36.79	66.39	267.10	47.68	104.81
	Transformer+CLIP	79.83	68.76	59.75	52.20	38.80	69.44	289.07	49.98	112.38
	Transformer+RSCLIP	79.28	68.46	59.89	52.78	**39.15**	**69.96**	290.22	**50.07**	**113.03**
	Transformer+CLIP+RN101	77.74	66.27	57.26	49.84	36.24	66.52	269.40	47.50	105.50
	Transformer+RSCLIP+RN101	**80.20**	**69.26**	**60.43**	**53.12**	38.54	69.24	**290.90**	49.85	112.95

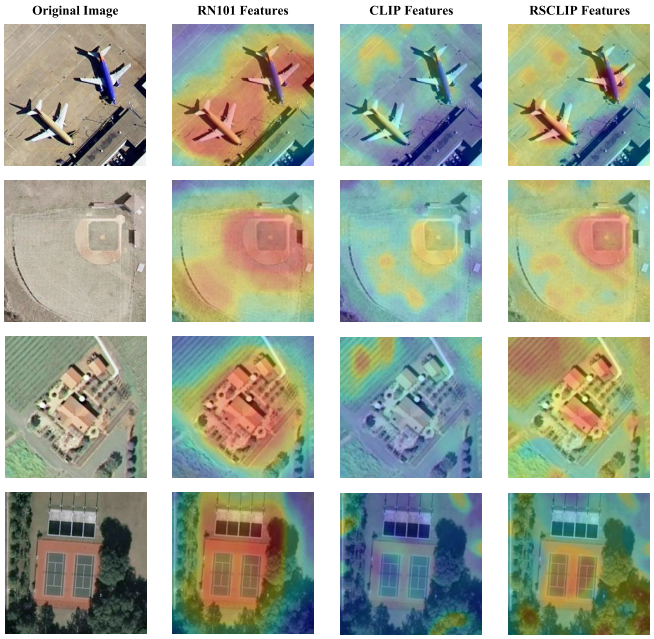


Fig. 5. Some samples of different visualized visual features. The RN101 features are extracted from the ResNet-101 model. The CLIP features are extracted from the original CLIP image encoder without fine-tuning, while the RSCLIP features are extracted from the image encoder of the fine-tuned model, RSCLIP. The warmer the color, the higher the attention value.

CLIP image encoder and the RSCLIP image encoder in this article.

Table VI shows the ablation experiments on different kinds of visual features. The difference between **"Transformer + CLIP"** and **"Transformer + RSCLIP"** is whether the original CLIP model has been fine-tuned with remote sensing data, resulting in the difference between the CLIP features and the RSCLIP features. Under the condition that the RSCLIP features and the ResNet-101 features contain the information of RSI in different views, it deserves to further explore their fusion. Compared with "Transformer + RSCLIP" or "Transformer + RN101," **"Transformer + RSCLIP + RN101"** denotes the Transformer with both the RSCLIP features and the ResNet-101 features, where the attention-based feature fusion module is applied to fuse these two kinds of features. As we have described in detail how we fuse the RSCLIP features and the ResNet-101 features via the attention-based feature fusion module in Section III-A, it is easy to understand the structure of "Transformer + RSCLIP + RN101." Besides, we further explore the fusion of the CLIP features and the ResNet-101 features using the attention-based feature fusion module, resulting in **"Transformer + CLIP + RN101."**

For three datasets, it is observed that the visual features extracted from the CLIP image encoder or the RSCLIP image encoder show better performance than those from ResNet-101. When the attention-based feature fusion module is applied to fuse the RSCLIP features and the ResNet-101 features, we discover that the overall performance can be further enhanced. Specifically, in the UCM-Captions dataset, "Transformer + RSCLIP + RN101" ranks first in seven-ninths of the evaluation metrics. Although the BLEU-1 score and the BLEU-2 score of "Transformer + RSCLIP + RN101" are a little worse than those of "Transformer + RSCLIP," the gap of scores is less than 0.20 points. In the Sydney-Captions dataset, "Transformer + RSCLIP + RN101" outperforms other baseline models on all metrics by an obvious margin, proving the excellence of the multi-view fused visual features. In the RSICD, "Transformer + RSCLIP + RN101" gains the best performance on the BLEU-n scores ($n = 1, 2, 3, 4$) and the CIDEr score, while the rest metric scores are slightly behind those of "Transformer + RSCLIP." It is worth noting that "Transformer + CLIP + RN101" outperforms "Transformer + RN101" overall, but falls significantly short of "Transformer + RSCLIP + RN101." This finding suggests that the RSCLIP features play an important role in feature fusion and the combination of the CLIP features and the ResNet-101 features may be the suboptimal choice. From the above results, it also concludes that the RSCLIP features may have strong RSI representations, which are beneficial to RSI captioning in comparison with the CLIP features or the ResNet-101 features.

In addition, we show some visualized results of different visual features as shown in Fig. 5. Our visualized results are

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TABLE VII
ABLATION RESULTS OF DIFFERENT PREFIX EMBEDDINGS ON THE UCM-CAPTIONS DATASET

Prefix Type	Method: Mapping Network	Visual Feature Extractor	*B1*	*B2*	*B3*	*B4*	*M*	*R*	*C*	*S*	S_m
Static Prefix	-	-	88.12	82.15	76.79	71.61	45.55	82.23	345.13	50.72	136.13
Dynamic Prefix	Linear Layer	MVFF	87.47	81.88	76.50	71.45	46.96	82.51	343.40	49.20	136.08
		Pretrained ResNet-101	85.31	79.19	74.28	69.76	44.29	79.37	337.41	49.98	132.71
		RSCLIP Image Encoder	85.19	79.49	74.52	70.03	46.12	80.98	332.34	49.12	132.37
	Adaptive Average Pooling	MVFF	87.25	81.78	77.08	72.73	46.23	81.72	357.84	47.38	139.63
		Pretrained ResNet-101	88.25	83.58	78.94	74.72	47.76	83.04	357.31	47.88	140.71
		RSCLIP Image Encoder	**89.05**	**84.21**	**79.9**	**75.89**	**48.97**	**84.41**	**379.05**	**52.78**	**147.08**

"-" means any mapping network and visual feature extractor is not used.

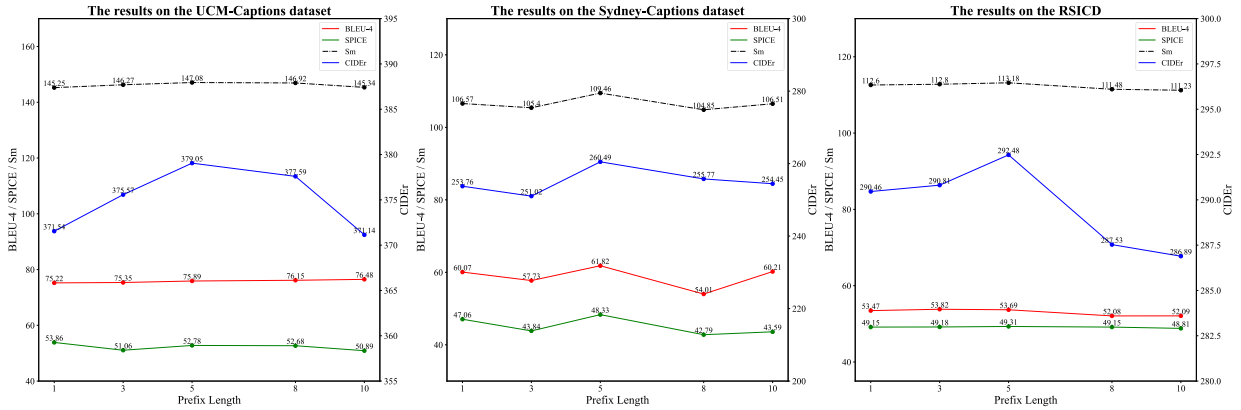


Fig. 6. Effects of the prefix length on the model performance over three datasets. For each prefix length, we report the BLEU-4 (red line), SPICE (green line), CIDEr (blue line), and S_m (black dashed line) scores.

based on cosine similarity between the global features and the local features. We consider that the global features contain more all-around semantic information, and the local features are related to the global features. Consequently, the higher the similarities between the global and local features, the more attention is paid to that local region. For the ResNet-101 features, we regard the output of the final convolution layer as the local features and the mean of the local features as the global features. Since both the original CLIP image encoder and the RSCLIP image encoder are Visual Transformer models, we consider the final output of the classification token as the global features and the final output of other tokens as the local features for both CLIP features and RSCLIP features. It is seen that the ResNet-101 features and the RSCLIP features can recognize the objects in RSIs, while the latter gives a more accurate range of the salient objects. Although the CLIP features also highlight some target objects, they perform less obviously compared to the RSCLIP features. To further analyze, we hold that the RSCLIP model learns more knowledge of image–sentence alignment and shows excellent performance than the original CLIP model.

2) Visual Prefix Analysis : Reviewing the introduction to visual prefix embeddings in Section III-B, the two-submodule framework consists of a visual feature extractor and a mapping network. To comprehensively analyze the effects of visual prefixes, extensive experiments are conducted including three core factors: the choice of visual feature extractor, the choice of mapping network, and the prefix length.

Table VII shows experiments on different combinations of visual feature extractors and mapping networks with identical settings on the UCM-Caption dataset, where an MVFF module and a BERT-based caption generator are both employed. To start with, we conclude two types of prefixes: dynamic prefixes and static prefixes. Visual prefix embeddings, produced by the combined networks, can be termed dynamic prefixes. Conversely, manually designed prefixes are referred to as static prefixes. In our experiments, we use the word embeddings of "the model translates the image into" as a static prefix. It is observed that the static prefix is still feasible and effective for our method.

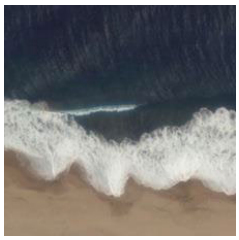
As shown in Fig. 3, different visual feature extractors and mapping networks are selected in our experiments. For the choice of visual feature extractor, we choose the pretrained ResNet-101 model, the RSCLIP image encoder, and the MVFF module, as shown in Table VII. The mapping network typically employs a linear layer and adaptive average pooling operation, which are effective for mapping feature vectors into the specified dimension vectors. Therefore, we conduct extensive experiments with various combinations of these choices. Table VII shows that the overall results of using adaptive average pooling outperform those of using a linear layer, regardless of any visual feature extractor. While linear layers can map the feature vectors into the target-dimension vectors, they may lead to the loss of critical information from the visual features extracted by the visual feature extractors, resulting in reduced performance. Adaptive average pooling is essentially a

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TABLE VIII
ABLATION RESULTS OF DESIGNED MODULES ON THE UCM-CAPTIONS DATASET

Method	MVFF	P	B	*B1*	*B2*	*B3*	*B4*	*M*	*R*	*C*	*S*	S_m
Transformer+RN101(baseline)	×	×	×	79.80	72.44	66.05	60.19	40.44	75.14	288.65	40.50	116.11
Transformer+MVFF	✓	×	×	88.05	82.23	77.65	73.58	47.94	82.79	362.59	51.94	141.73
Transformer+MVFF+Prefix	✓	✓	×	87.62	82.72	78.17	73.96	46.97	82.18	352.56	49.57	138.92
Transformer+MVFF+BERT	✓	×	✓	87.84	83.08	78.74	74.67	**49.06**	84.31	360.37	52.30	142.10
MVP	✓	✓	✓	**89.05**	**84.21**	**79.90**	**75.89**	48.97	**84.41**	**379.05**	**52.78**	**147.08**

"MVFF", "P", and "B" refer to the Multi-View Feature Fusion module, the visual Prefix mapping module, and the Bert-based caption generator, respectively.



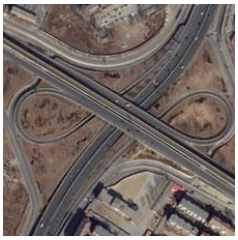
GT: White waves in ocean is near a beach.
Baseline: Yellow beach is near a piece of green ocean.
Ours: White waves in green ocean are near yellow beach.

(a)



GT: A building with a swimming pool is surrounded by many green trees.
Baseline: Many green trees are around a swimming pool.
Ours: Many green trees are around a building with a swimming pool.

(b)



GT: Some buildings and bareland are near a viaduct .
Baseline: Some buildings and green trees are near a viaduct.
Ours: Many buildings and bareland are near a viaduct.

(c)



GT: Some houses arranged neatly with plants surrounded in the medium residential area .
Baseline: A medium residential area with houses and trees.
Ours: A medium residential area with some houses arranged neatly.

(d)



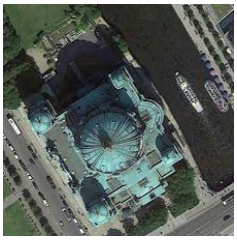
GT: A playground is surrounded by some green trees and several buildings.
Baseline: Some green trees are around a playground.
Ours: Several buildings and some green trees are around a playground.

(e)



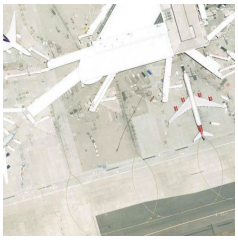
GT: There are two white straight freeways closed together with cars on them.
Baseline: There are some cars are on the freeways.
Ours: Two straight freeways closed to each other with some cars on them.

(f)



GT: Several buildings and green trees are around a church .
Baseline: Some buildings are around a church.
Ours: Some buildings and some green trees are around a church.

(g)



GT: A white airplane parked on the airport with some runways passed by.
Baseline: A big airport with only a white airport parked on.
Ours: There is a white airplane parked on the airport with a runway beside.

(h)

Fig. 7. Some captioning examples on three datasets. The **GT** denotes the ground-truth sentence. The **Baseline** denotes the sentence generated by the baseline model, "Transformer+RN101." The **Ours** denotes the sentence generated by our method. The words in red are incorrectly matched with the corresponding image, while the words in blue describe the image appropriately. (a) Wave. (b) Swimming pool. (c) Bareland. (d) Medium residential area. (e) Playground. (f) Freeway. (g) Church. (h) Runway.

downsampling operation, which is beneficial to aggregate key information in the extracted visual features. Moreover, we find that when adopting adaptive average pooling, the results using the RSCLIP image encoder outperform those using the MVFF module or the pretrained ResNet-101 model. Further analysis shows that the visual features extracted from the RSCLIP image encoder contain more sentence-level information, probably producing better visual prefix embeddings. The effects of the features extracted from the MVFF module and the pretrained ResNet-101 model on the visual prefix embedding are not readily apparent. The results also indicate that the combination of the RSCLIP image encoder with adaptive average pooling is considered the most effective choice in our method, and this combination is consistently used in our experiments on three datasets.

The prefix length is another important parameter that can enhance the model's performance with certain values [15], [18]. Fig. 6 presents the experiments of different prefix lengths on three datasets. When increasing the size of the prefix length, it is evident from the figure that all metrics initially increase and then slightly decrease. For all three datasets, the results of the prefix length of five outperform those of other prefix lengths in terms of the overall performance. These results indicate that the size of the prefix length affects the performance of the model. To further analyze, we suggest that the size of prefix length may be influenced by factors such as the dataset size or the length of the ground-truth sentences.

3) Effects of Different Designed Modules: Table VIII presents ablation experiments on designed modules on the UCM-Captions dataset. We denote "Transformer + RN101" as the baseline model. Besides, "MVFF," "P," and "B" in the table are short for the multi-view feature fusion module, the visual prefix mapping module, and the Bert-based caption generator, respectively. Specifically, the performance

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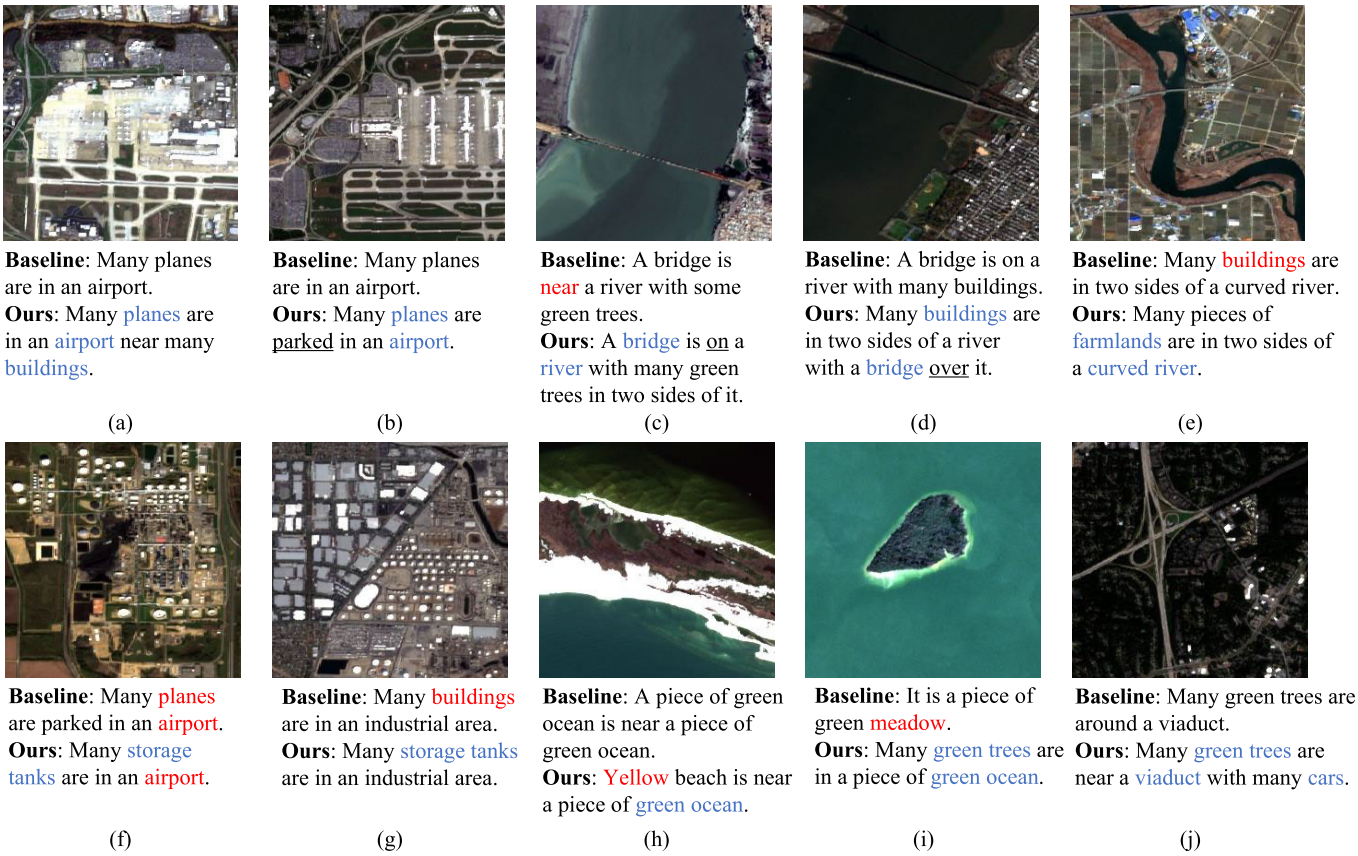


Fig. 8. Some captioning examples on the Sentinel-2 images. The **Baseline** denotes the sentence generated by the baseline model, "Transformer + RN101." The **Ours** denotes the sentence generated by our method. The words in red are incorrectly matched with the corresponding image, while the words in blue describe the image appropriately. The underlined words highlight the accurate details of the corresponding image. (a) Airport. (b) Airport. (c) Bridge. (d) Bridge. (e) Farmland. (f) Storage tank. (g) Storage tank. (h) Ocean. (i) Ocean. (j) Viaduct.

of three variants of the proposed method surpasses that of "Transformer + RN101." "Transformer + MVFF + BERT" outperforms "Transformer + MVFF" by 1.48% on BLEU-4, 2.34% on METEOR, and 1.84% on ROUGE_L. Additionally, "Transformer + MVFF + Prefix" slightly surpasses "Transformer + MVFF" on BLEU-2, BLEU-3, and BLEU-4. Our MVP model, which includes "MVFF," "P," and "B," exceeds other compared models by a significant margin on nearly all metrics. This indicates the effectiveness of the combination of our designed modules. Overall, these results demonstrate that each module can enhance the model's performance to some extent, and when all modules are combined, they can further improve the overall performance of the entire model.

F. Qualitative Analysis

As discussed and analyzed above, extensive quantitative results strongly prove the superiority of our method. To demonstrate the effectiveness of the proposed method, we show some qualitative results on the test set of three datasets in Fig. 7. In the figure, the "GT" denotes the ground-truth sentences of the image, and the "Ours" denotes the sentences generated by our method. Particularly, the "Transformer + RN101" model serves as the baseline model and is denoted as "Baseline." Red words and blue words are used to mark the incorrect and well-described contents about the corresponding image, respectively.

Specifically, it is observed that the descriptions of our method are more accurate and comprehensive than those of the baseline model. For example, the generated sentences of our method can describe more salient objects [such as "waves" in Fig. 7(a) and "building" in Fig. 7(b)] in detail. The previous methods also meet a problem that generated sentences are easily affected by some frequent coexistence objects, like "buildings" and "trees." However, our method can alleviate this problem to acquire better descriptions. As shown in Fig. 7(c), "green trees" in the baseline sentence do not exist in the corresponding images, while "bareland" is correctly described in the sentence generated by our method. While our method achieves overall superior performance, it has limitations in accurately describing the number of target objects, as shown in Fig. 7(h).

To further estimate our method in real-world scenarios, we test the proposed model on the Sentinel-2 images from the SEN1-2 dataset. For inference, we only use the optical images from the Sentinel-2 satellite. Fig. 8 displays the results obtained from our method compared to the baseline model "Transformer + RN101" on the Sentinel-2 images. Red words refer to improper contents, blue words are used to mark the well-described objects, and the underlined words highlight the accurate details of the corresponding image. In particular, both our method and the baseline model are trained on the RSICD and subsequently tested directly on the Sentinel-2

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TABLE IX
COMPARISON RESULTS WITH STATE-OF-THE-ART METHODS ON THE NUMBER OF PARAMETERS, SPE (SECOND PER EPOCH), TRAINING TIME, AND INFERENCE SPEED (IMAGES PER SECOND). ALL RESULTS ARE REPORTED BASED ON THE RSICD

Method	Parameters↓	SPE↓	Training time↓	Inference speed↑
Up-Down [48]	45.18M	-	266min	16.73
RASG [6]	53.44M	-	228min	16.50
MVP	98.23M	150.42s	**116min**	**30.17**

"↑" indicates that a higher related metric score is better while "↓" indicates that the lower one is better.
"-" means the number of SPE is not reported in the reference papers, so the value of SPE will be omitted correspondingly.

images. The reason for using models trained on the RSICD is that Sentinel-2 images share a similar style with RSICD images, allowing the generalization ability of the trained models to be tested to some extent. Additionally, the RSICD contains more object classes and images from across the globe, potentially enabling these trained models to capture more RSI information.

Our method can generate descriptions in most scenes, which are new to the model. Compared with the baseline model, the generated sentences of our method provide descriptions with more accurate objects such as "farmlands" in Fig. 8(e) and "storage tanks" in Fig. 8(g). Although mistakes also appear due to the highly similar objects, our method offers advantages in terms of describing salient objects and details. Based on the above analysis, our method exhibits a certain degree of knowledge transferability and the potential for practical application in real-world scenarios.

G. Speed Performance

Efficient model design is also a crucial aspect of the study. In this part, we report the number of model parameters, SPE (second per epoch), training time, and inference speed (images per second) of our method. Following the results in [6], Table IX represents the comparison results with other SOTA methods on the RSICD. Specifically, our method offers clear advantages in terms of training time and inference speed, with an almost 50% improvement compared to other methods. This is because we locally extract and store the visual features of RSIs before training the model, and then directly load them from the disk during the training and inference stage. Therefore, there is no need for additional computation time for feature extraction. Compared with the SOTA methods, our method increases the number of model parameters due to the utilization of a multilayer attention-based feature fusion module and BERT-based caption generator. Powered by large and complex modules, our method achieves superior performance while using more parameters.

V. CONCLUSION

In this article, we propose a novel method for RSI captioning to enhance RSI representations and generate high-quality sentences. The MVP model contains three designed modules: an MVFF module, a visual prefix mapping module, and a BERT-based caption generator. First, the attention-based MVFF module is designed to fuse multi-view features for better RSI representations. Then, to boost sentence generation, the flexible visual prefix mapping module is proposed to produce visual prefixes from the image. Finally, the BERT-based caption generator combines the two mentioned modules and generates sentences for the corresponding image. Extensive experiments indicate that our method surpasses current SOTA methods on the RSICD, the UCM-Caption dataset, and the Sydney-Captions dataset. The ablation results demonstrate the individual superiority of each module, and their combined use further improves the overall model performance.

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